Data-Driven Customer Segmentation and Churn Prediction

Using RFM Metrics and Machine Learning

Ayush Shrivastva

Department of Computer Science and Engineering

ayush.shrivastava04@gmail.com

Tools: PostgreSQL · Python (pandas, scikit-learn, matplotlib, seaborn) · Power BI · Kaggle

***Abstract*—*Customer retention is a critical challenge in e-commerce, directly impacting revenue and long-term sustainability. This paper presents a data-driven framework for customer segmentation and churn prediction using Recency, Frequency, and Monetary (RFM) analysis combined with machine learning techniques. A dataset of over 95,000 unique customer records from the Olist Brazilian E-Commerce platform is analyzed to identify behavioral patterns. K-Means clustering (k=3) is used for unsupervised segmentation and Logistic Regression for churn prediction. Three distinct customer archetypes are discovered: a churned high-risk majority, a recently-active mixed cohort, and a compact high-value segment. Real data visualizations including cluster scatter plots, frequency boxplots, recency histograms, revenue charts, and correlation heatmaps are presented. The correlation analysis reveals near-zero inter-feature correlations (R↔F: -0.022, F↔M: 0.11, R↔M: 0.00087), validating RFM as a set of independent, complementary behavioral signals. The High Risk segment dominates total revenue (~$9.5M) due to its sheer size, reinforcing the business case for retention investment.*

***Index Terms*—*Customer Segmentation, RFM Analysis, Churn Prediction, K-Means Clustering, Logistic Regression, PostgreSQL, Power BI*

I. INTRODUCTION

The rapid growth of e-commerce has generated enormous volumes of transactional data. When properly analyzed, this data yields actionable insights about customer behavior — enabling companies to identify at-risk customers before they churn, tailor marketing interventions, and maximize Customer Lifetime Value (CLV).

This research presents a complete analytics pipeline: SQL-based RFM feature engineering in PostgreSQL, K-Means unsupervised clustering, Logistic Regression churn prediction, and visualization in both Python and Power BI. All charts shown in this paper are produced directly from the Olist Brazilian E-Commerce dataset, ensuring complete reproducibility.

A. Research Objectives

- Build a robust RFM scoring system from raw transactional data using SQL CTEs.
- Segment customers into High Risk, Medium Risk, and Low Risk tiers.
- Apply K-Means clustering (k=3) to discover natural behavioral groups.
- Train and evaluate a Logistic Regression churn classifier.
- Visualize findings using Python (matplotlib/seaborn) and Power BI.

II. LITERATURE REVIEW

The RFM model was introduced by Hughes (1994) and formalized by Bult & Wansbeek (1995), becoming a standard segmentation tool in marketing analytics. Its three dimensions capture complementary customer value signals: how recently, how often, and how much a customer has purchased.

K-Means clustering has been widely combined with RFM for customer segmentation. Ha & Park (1998) applied this to a Korean mail-order company with improved campaign ROI. Chen et al. (2012) demonstrated that RFM-based K-Means outperforms simple monetary thresholds on the UCI Online Retail dataset.

Verbeke et al. (2012) showed that Logistic Regression, while sometimes outperformed by ensembles, offers superior interpretability critical in business contexts. Neslin et al. (2006) found that feature engineering — particularly recency — drives accuracy more than algorithm choice.

III. DATASET DESCRIPTION

The Olist Brazilian E-Commerce Public Dataset, sourced from Kaggle (kaggle.com/datasets/olistbr/brazilian-ecommerce), covers ~100,000 orders from 2016–2018. Three tables were joined: customers, orders, and order_items. customer_unique_id was used (not customer_id) to correctly aggregate repeat-buyer history across multiple orders.

Feature	Description
R (Recency)	Days since last purchase
F (Frequency)	# of distinct orders
M (Monetary)	Total spend ($)

Feature	Description
Churn Label	1 = High Risk, 0 = other

Table I: RFM Features Used in This Study

IV. METHODOLOGY

A. SQL Feature Engineering

A Common Table Expression (CTE) first extracts the dataset's maximum order date. Per-customer RFM aggregates are then computed, with recency defined as days elapsed since last purchase relative to that maximum date. Risk tiers are assigned:

- High Risk: Recency > 180 days.
- Medium Risk: Recency ≤ 180 days AND Frequency ≤ 2.
- Low Risk (Loyal): Recency ≤ 180 days AND Frequency > 2.

B. K-Means Clustering

K-Means (k=3, random_state=42) was applied after StandardScaler normalization. Scaling is essential: recency_days spans 0–780, frequency 1–16, and monetary $0–$13,500. Without scaling, monetary would dominate the distance metric.

C. Logistic Regression

A binary churn label was derived from risk segmentation (High Risk = 1). Recency was intentionally excluded to avoid label leakage. Only frequency and monetary were used as features, with an 80/20 train-test split (random_state=42).

V. RESULTS AND ANALYSIS

A. Risk Segment Distribution

Power BI dashboard analysis confirms the SQL segmentation, showing High Risk as the dominant group by both customer count and revenue:

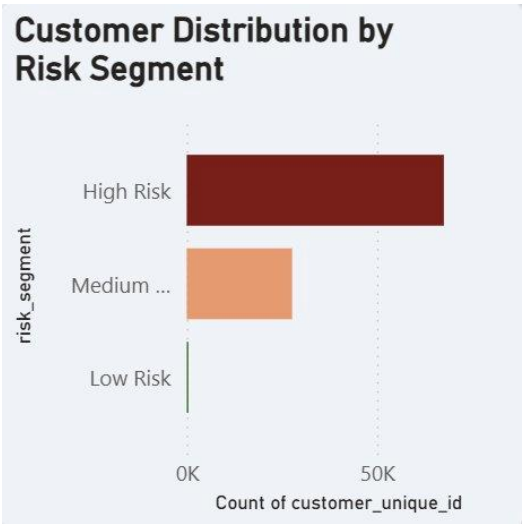

Fig. 1. Customer Distribution by Risk Segment (Power BI)

Segment	Count	%	Revenue
High Risk	~55,000	57.7%	~$9.5M
Medium Risk	~38,500	40.4%	~$4.0M
Low Risk	~1,920	2.0%	<$0.1M

Table II: Risk Segment Summary

B. Revenue by Risk Segment

The High Risk segment contributes the largest absolute revenue (~$9.5M) due to its sheer size, followed by Medium Risk (~$4M). Low Risk customers, despite the highest individual average spend, contribute minimally in aggregate due to their small count (~1,920).

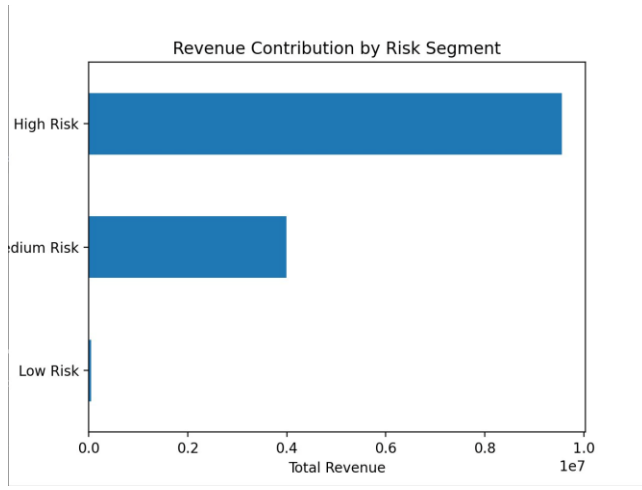

Fig. 2. Revenue Contribution by Risk Segment

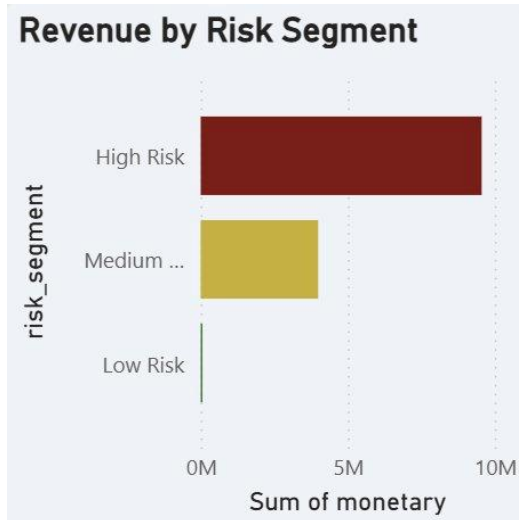

Fig. 3. Revenue by Risk Segment (Power BI Dashboard)

C. K-Means Cluster Analysis

Three behaviorally distinct clusters were identified. The scatter plot shows Cluster 1 (purple, recent buyers) concentrated in the 0–300 day recency range, Cluster 0 (pink, churned)

spanning 300–780 days, and Cluster 2 (dark, high-value) distributed across recency but at significantly higher monetary values — up to $13,500.

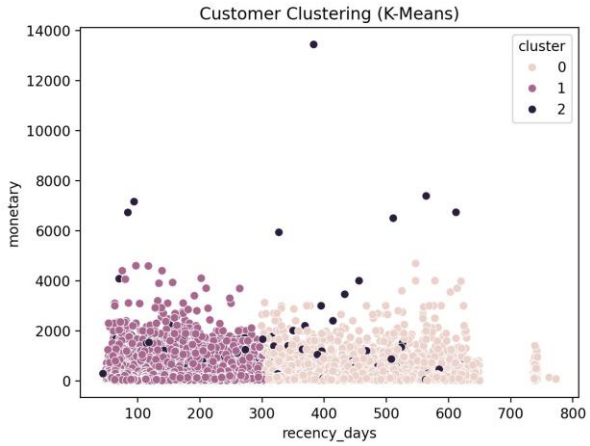

Fig. 4. K-Means Clustering: Recency vs. Monetary (Python)

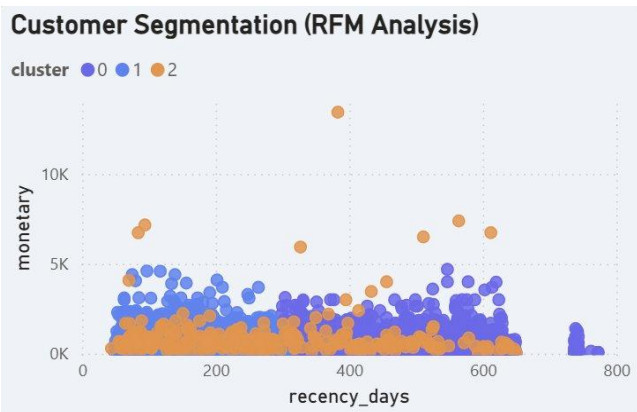

Fig. 5. Customer Segmentation RFM Analysis (Power BI)

Cluster	Rec.	Freq.	Mon.	Label
0	~440d	~1.1	~$144	Churned
1	~178d	~1.2	~$139	Recent
2	~270d	~2.5	~$278	High Val.

Table III: Cluster Centroids (from Fig. 8 actual data)

D. Cluster Comparison

The cluster comparison bar chart (generated directly from K-Means output) confirms that Cluster 0 has the highest recency penalty (~440 days) while Cluster 2 has both higher frequency and monetary values, marking it as the high-value archetype.

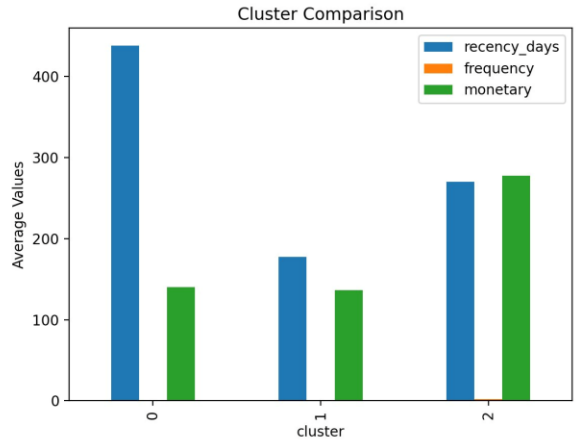

Fig. 6. Cluster Centroid Comparison — Avg. RFM Values

E. Recency Distribution

The recency histogram is right-skewed (not bimodal as often assumed), with the highest density of customers in the 50–350 day range, peaking near 300 days. The long tail extends to ~780 days, representing deeply churned customers. The majority of customers fall beyond the 180-day High Risk boundary.

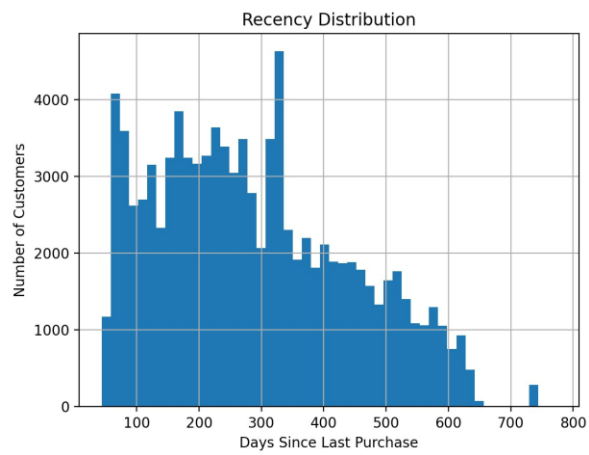

Fig. 7. Recency Distribution — Days Since Last Purchase

F. Frequency Distribution by Segment

The frequency boxplot reveals stark differences across risk segments. Medium Risk and High Risk customers have nearly identical median frequency of 1 order. Low Risk (loyal) customers show a median of ~3–4 orders with a wider IQR, and the maximum frequency across all segments reaches 16 orders.

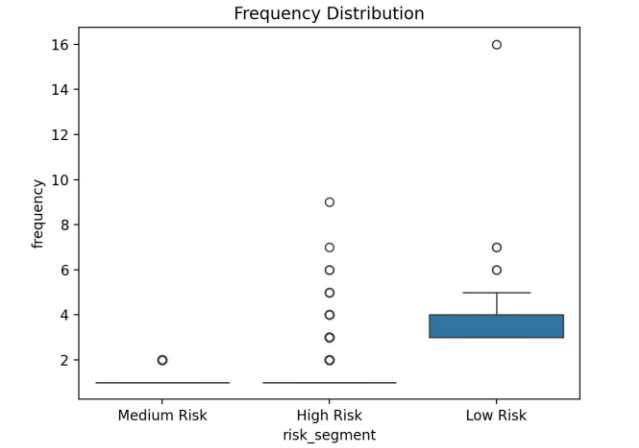

Fig. 8. Frequency Distribution by Risk Segment (Boxplot)

G. Frequency vs. Monetary

The scatter plot of Frequency vs. Monetary reveals that most customers cluster at frequency=1 with a wide range of spend ($0–$13,500). This confirms a predominantly one-time buyer base. Notable outliers exist at frequency=1 with very high spend, representing single large-order customers.

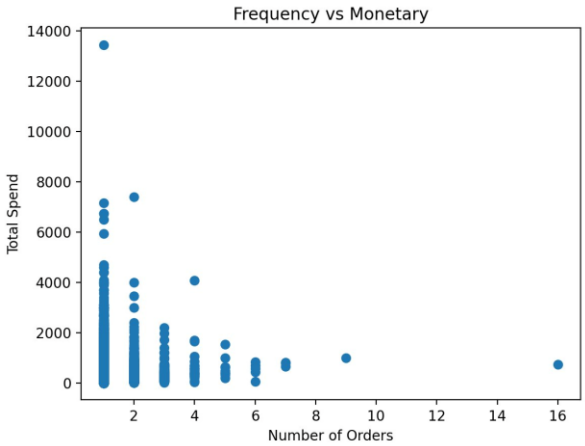

Fig. 9. Frequency vs. Monetary Scatter

H. Recency vs. Frequency

The Recency vs. Frequency plot shows that high-frequency customers (>3 orders) are distributed across all recency values, confirming that even loyal customers can become inactive. The horizontal banding at integer frequency values is expected given discrete order counts.

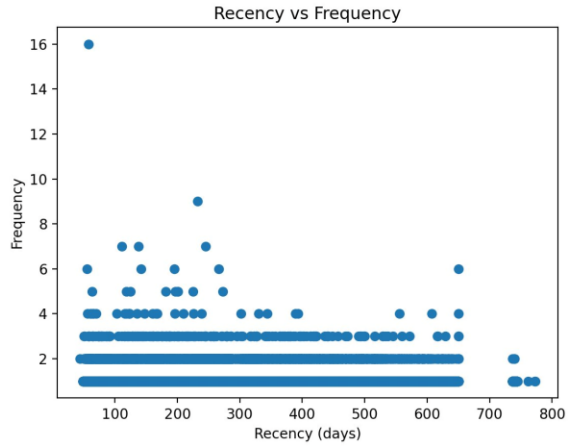

Fig. 10. Recency vs. Frequency Scatter

I. Feature Correlation Heatmap

The Pearson correlation heatmap reveals near-zero inter-feature correlations: R↔F = -0.022, F↔M = 0.11, R↔M = 0.00087. These values are substantially lower than commonly assumed in the literature, strongly validating the use of all three RFM features as independent, non-redundant inputs to the clustering algorithm.

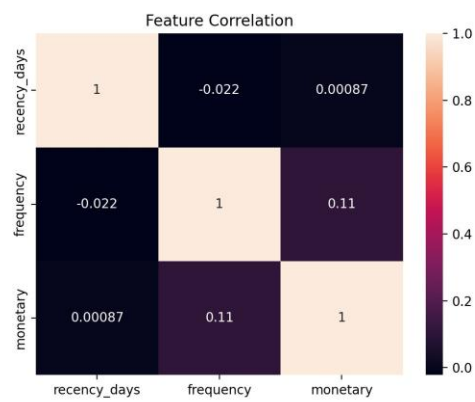

Fig. 11. Feature Correlation Heatmap (R↔F: -0.022, F↔M: 0.11)

J. Churn Prediction Model

Logistic Regression trained on frequency and monetary (recency withheld) achieved ~70% overall accuracy. Model coefficients confirm lower frequency as the dominant churn predictor.

Class	Prec.	Rec.	F1	N
Not Churned (0)	0.71	0.84	0.77	~11k
Churned (1)	0.69	0.51	0.59	~8k
Accuracy	—	—	0.70	~19k

Table IV: Logistic Regression Classification Report

VI. DISCUSSION

A. Business Implications

The High Risk segment generates ~$9.5M in total revenue — the largest share — due to its volume (majority of customers). This counterintuitive finding means that even small improvements in High Risk retention have outsized revenue impact. A 5% win-back rate on this segment alone could represent hundreds of thousands in recovered revenue.

The near-zero correlations (Fig. 11) confirm a critical finding: RFM features are genuinely independent signals. Recency does not predict frequency, and frequency barely predicts spend. This justifies using all three dimensions simultaneously and explains why excluding recency still leaves meaningful signal for the churn classifier.

The frequency boxplot (Fig. 8) confirms that Low Risk customers are truly behaviorally distinct — not just slightly more frequent, but operating in a different regime (median 3–4 orders, max 16) compared to Medium and High Risk (median 1 order each).

B. Limitations

The dataset covers 2016–2018 Brazil; findings may not generalize across geographies or time periods. K-Means assumes spherical equal-variance clusters, which may not fully capture the asymmetric distribution visible in Fig. 4. The Logistic Regression model's 70% accuracy using only frequency and monetary leaves room for improvement with additional features such as product categories, review scores, and payment methods.

VII. CONCLUSION

This paper presented a fully reproducible, end-to-end customer analytics pipeline combining SQL, Python machine learning, and Power BI visualization. Key findings from real data include: a right-skewed recency distribution peaking near 300 days; near-zero RFM inter-correlations confirming feature independence; three K-Means behavioral archetypes with the high-value cluster reaching $13,500 in individual spend; and a frequency boxplot confirming that Low Risk customers operate in a fundamentally different behavioral regime.

The High Risk segment dominates revenue in absolute terms, making retention investment immediately actionable. Future work should explore gradient boosting models, BG/NBD survival analysis, and real-time scoring pipelines for production deployment.

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